

How to Publish in Q1 Journals and Top-Tier Conferences?

Comprehensive step-by-step guideline for preparing and publishing research

Lama Ayash

July 2026

Day 1

RESEARCH

What is Research?

Research is the **systematic** and scientific investigation of a **specific problem** to generate **new knowledge**, **evidence**, or **solutions**.

What is Research?

Rr

Again

Search

Find

Process

Step by Step

What is NOT Research?

The following activities are not research by themselves:

- Experimenting with technology without a clear research question
- Developing code only to build a working application
- Deploying an existing commercial or standard solution
- Repeating previous work without a new objective, analysis, or contribution



However, each of these can be done as part of research

The Three Approaches to Research

Quantitative Research

Uses numerical data, measurements, and statistical analysis to test hypotheses and validate claims.

Qualitative Research

Uses observations, interviews, and experiences to understand behaviors, perceptions, and patterns.

Mixed-Methods Research

Combines quantitative and qualitative approaches

RESEARCH PROBLEM

Research Problem

A research problem is a clear statement describing:

- A **gap** in existing knowledge.
- A **challenge** that needs investigation.
- A **contradiction** in previous findings.
- A **real-world issue** requiring a better solution.

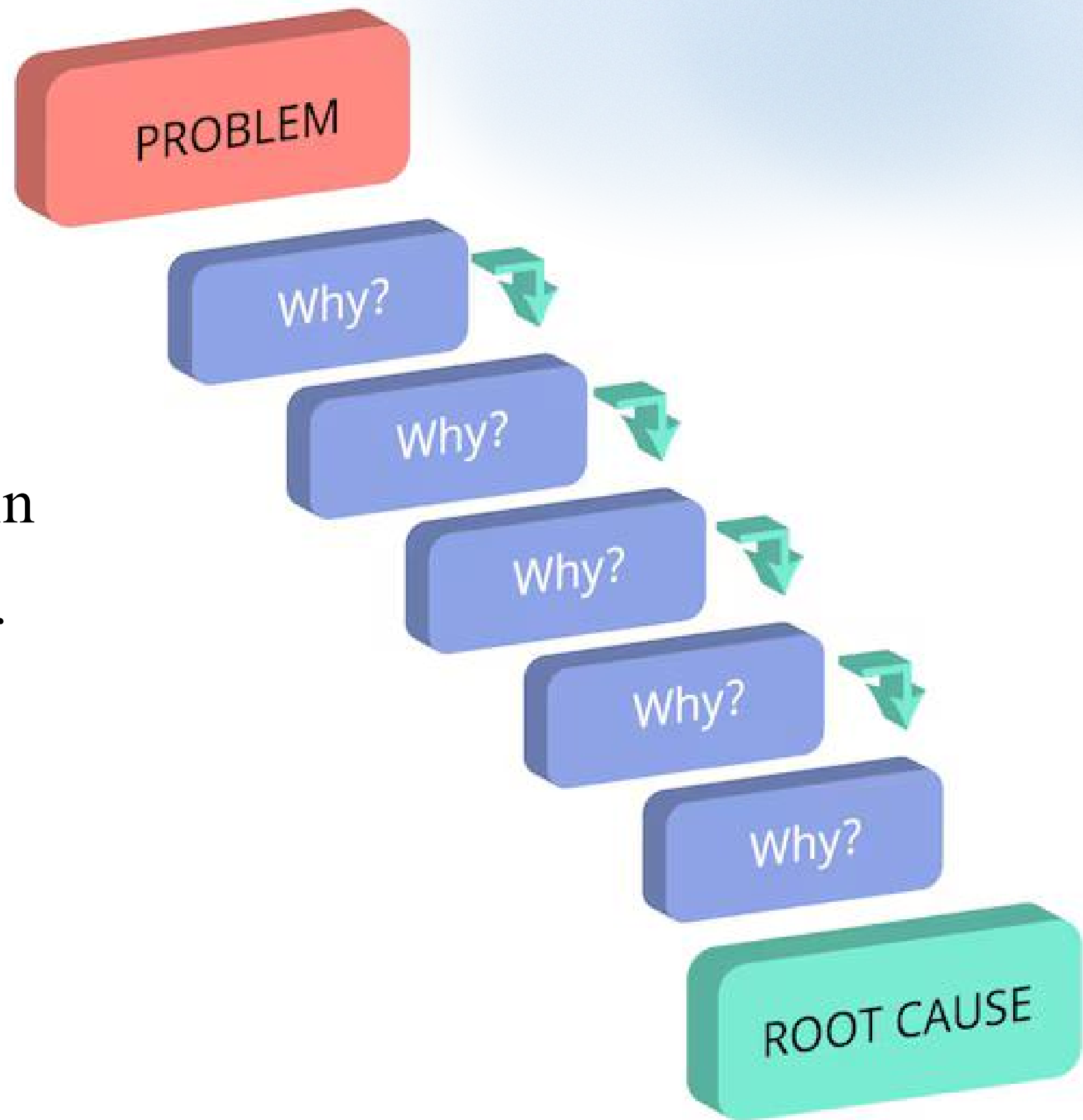
Research Problem

The research problem gives the study a clear direction by helping researchers define:

- The **aim** of the study
- The main **research question**
- The key concepts and variables
- The **scope** of the investigation

Research Problem

- The Seven Whys can be used in research to identify the deeper cause behind a problem.



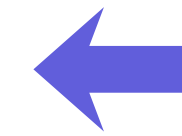
Research Problem

Problem 1: The NLP model performs poorly on Arabic dialects

Problem 2: The robot loses localization accuracy inside tunnels

Research Problem

Diabetes is a major global health problem, affecting approximately 830 million people worldwide in 2022 and directly causing 1.6 million deaths in 2021. It can also result in severe complications, including cardiovascular disease, kidney failure, lower-limb amputation, and permanent vision loss [1].



Problem background

Research Problem

Diabetes is a major global health problem, affecting approximately 830 million people worldwide in 2022 and directly causing 1.6 million deaths in 2021. It can also result in severe complications, including cardiovascular disease, kidney failure, lower-limb amputation, and permanent vision loss [1].

Although artificial intelligence models have achieved high accuracy in detecting diabetic retinopathy from retinal images, their performance may decline when applied to different hospitals, patient populations, imaging devices, or image-quality conditions. Many models also lack sufficient external validation and interpretability, which limits their reliability and adoption in real-world clinical practice.

 **Problem background**

 **Technical Problem**

Research Problem

Current AI models may struggle to generalize across hospitals, patients, and imaging devices. Their performance can also decline with low-quality images, while limited interpretability and insufficient external validation reduce their reliability in clinical practice.

 **Sub-problems**

Research Problem

Current AI models may struggle to generalize across hospitals, patients, and imaging devices. Their performance can also decline with low-quality images, while limited interpretability and insufficient external validation reduce their reliability in clinical practice.

← **Sub-problems**

There is still a need for an AI model that can detect diabetic retinopathy accurately across diverse patients, hospitals, imaging devices, and image-quality conditions while providing interpretable and clinically reliable predictions.

← **Research gap**

Research Problem

☐	Background and Significance	How serious or important is the problem?	Introduces the topic, explains its impact, and shows why the problem deserves investigation.
☐	Technical Problem	What is wrong with current methods, systems, or models?	Defines the main technical limitation that prevents effective performance or application.
☐	Sub-Problems	What specific limitations contribute to the technical problem?	Breaks the main problem into smaller, clear, and measurable challenges.
☐	Research Gap	What has not yet been adequately solved or investigated?	Identifies what is missing, insufficient, inconsistent, or unexplored in existing research.

Research Problem

Common Mistakes:

- ✗ Choosing a problem that is too broad or vague
- ✗ Defining a problem that cannot be measured, tested, or investigated
- ✗ Proposing a solution

RESEARCH QUESTION

Research Question

A research question converts the research problem into a clear, focused, and answerable question that guides the study.

- What is the effect of X on Y?
- How can X be improved under Y conditions?
- What factors contribute to X?
- How does Method A compare with Method B?

Research Question

☐	Question Starter	How, what, why, can, does, or to what extent
☐	Method or Factor	The model, technique, intervention, or variable being studied
☐	Measurable Outcome	The result that will be evaluated
☐	Population or Context	The people, data, environment, or setting being studied

Example

- How does fine-tuning a transformer-based model affect sentiment-classification accuracy for Arabic dialects with limited annotated data?

Research Question

Common Mistakes:

- ✘ The question is too broad, vague, or includes several problems at once.
- ✘ The outcome cannot be clearly measured or evaluated.
- ✘ The question is not feasible or does not directly address the research gap.

AIM AND OBJECTIVES

Research Aim

The research aim is a broad statement that defines the overall purpose of the study and what the researcher intends to achieve.

- Clear and focused
- Broad but achievable
- Directly connected to the research problem
- Free from unnecessary methodological details

Research Aim

Example

Robotics has become increasingly important across sectors such as manufacturing, healthcare, transportation, and service industries due to its potential to improve efficiency, safety, and automation. However, robotic systems continue to face challenges when operating in dynamic, unstructured, and unfamiliar environments. Therefore, the aim of this research is to develop an AI-enabled robotic system capable of accurately perceiving its surroundings, making reliable decisions, and adapting effectively to changing real-world conditions.

Research objectives

Research objectives translate the overall research aim into specific, measurable, and achievable tasks that guide the study

- Begin with an action verb
- Address one task or outcome
- Be measurable and realistic
- Directly support the research aim

Research objectives

SMART Principles for a Research Question:

☐	Specific	Clearly defines the research problem, population, variables, and context.
☐	Measurable	Can be investigated using data, evidence, experiments, or clearly defined evaluation metrics.
☐	Achievable	Can be completed using the available time, data, tools, expertise, and computational resources.
☐	Relevant	Addresses an important scientific, technical, or practical problem and contributes to the research field.
☐	Time-bound	Can be investigated and completed within a clearly defined research period.

Research objectives

Common Aim Starters

- To develop...
- To investigate...
- To evaluate...
- To explore...
- To improve...

Example

- To evaluate the accuracy and response time of an AI-based robot for object detection in dynamic indoor environments.
- To develop and assess a machine-learning model for detecting malicious network traffic while reducing false-positive rates.
- To develop an AI-based system that improves early diagnosis of skin cancer.

Research objectives

Research Aim	Research Objectives
A broad statement describing the overall purpose of the research.	Specific statements describing the steps required to achieve the research aim.
Emphasizes the main purpose or overall direction of the study.	Emphasizes measurable and actionable research tasks.
High-level and general.	Detailed, practical, and specific.
Usually one main aim.	Usually multiple objectives, commonly between two and five.
To investigate, to explore, to develop, to improve.	To identify, to design, to compare, to evaluate, to validate.

Activity 1

Join your project group and prepare the following:

- A problem statement of 3–5 sentences
- Two research questions
- One research aim
- Two research objectives



Duration: 10 minutes

LITERATURE REVIEW

Literature Review

A literature review is a **critical** summary and **analysis** of existing research related to a specific topic. It identifies major findings, research trends, methodological approaches, contradictions, and unresolved gaps in the field.



Literature Review

☐	Start with a review paper to find relevant studies,
☐	Select up-to-date research that directly relates to the research problem and objectives.
☐	Prioritize reputable peer-reviewed journals and leading conferences in the field.
☐	Synthesize results across multiple studies rather than describing each study separately.
☐	Discuss weaknesses, contradictions, unanswered questions, and areas that require further investigation.
☐	Explain how the existing literature supports, motivates, or distinguishes the proposed research.

Literature Review

Questions to Answer When Reading and Analyzing Research Papers

What is known?	What have previous studies already established?	Main findings, successful methods, important theories, and commonly used datasets
What remains uncertain?	Where do studies disagree or provide limited evidence?	Contradictory findings, methodological limitations, and unanswered questions
What is missing?	What has not yet been sufficiently investigated?	Understudied populations, conditions, methods, datasets, or applications
Why is further research needed?	Why is the identified gap important?	Scientific significance, practical impact, and potential benefits of addressing the gap

Literature Review

Common Mistakes

- ✗ Listing studies without comparing or connecting them
- ✗ Summarizing papers without critical analysis
- ✗ Ignoring contradictory or negative findings
- ✗ Relying on outdated or low-quality sources
- ✗ Failing to link the review to the research problem and question

Literature Review

Example

Recent advancements in MLTC have addressed various aspects of text and label modeling, yet critical gaps remain.

A number of studies have centered on enhancing semantic text representation, utilizing powerful transformer architectures. For instance, Liu et al. [9] and Zhang et al. [10] leverage deep contextual embeddings to improve textual understanding. However, these methods treat labels as mere tags, neglecting the semantic richness or ambiguity inherent in label descriptions.

Other works have shifted focus toward label co-occurrence modeling, constructing label graphs to capture inter-label dependencies. Models such as MAGNET [11] and LDGN [12] exploit graph attention mechanisms and dual-graph structures to represent label relations more effectively. Nevertheless, these approaches do not utilize expanded

Studies

Limitation

Studies

Limitation

Type of Reviews

1. Narrative Literature Review

A narrative literature review provides a broad and critical synthesis of existing research on a topic. It summarizes major theories, findings, trends, and research gaps without following the highly structured procedures used in systematic reviews.

Type of Reviews

1. Narrative Literature Review

Best suited for:

- Providing an overview of a broad research field
- Comparing different theories or perspectives
- Identifying general trends and emerging research directions

Type of Reviews

1. Narrative Literature Review

INTERACTIVE JOURNAL OF MEDICAL RESEARCH

Chustecki

Review

Benefits and Risks of AI in Health Care: Narrative Review

Margaret Chustecki, MBA, MD

Department of Internal Medicine, Yale School of Medicine, New Haven, CT, United States

Corresponding Author:

Margaret Chustecki, MBA, MD

Department of Internal Medicine

Yale School of Medicine

1952 Whitney Ave

3rd Floor

New Haven, CT, 06510

United States

Phone: 1 2038091700

Email: margaret.chustecki@imgnh.com



Type of Reviews

2- Systematic Literature Review

A systematic literature review uses a transparent and structured process to search for, select, evaluate, and synthesize relevant research studies. It follows predefined inclusion and exclusion criteria to reduce bias and improve reproducibility.

➤ Answering a focused research question using a rigorous, evidence-based analysis.

Type of Reviews

2- Systematic Literature Review

REVIEW

Open Access



Advancements in feature selection and extraction methods for text mining: a review

Lama Ayash^{1*}, Abdulmohsen Algarni¹ and Omar Alqahtani¹

*Correspondence:

Lama Ayash

444800100@kku.edu.sa

¹Department of Computer Science,
King Khaled University, Alfaraa,
Abha 61421, Aseer, Saudi Arabia

Abstract

Textual data are among the most popular data types that provide useful knowledge today. The need to extract insights from text has initiated and overseen the progression of research in the text mining domain. The challenge of addressing textual data comes from the complexity of text processing. The nature of text presents challenges that demand sophisticated processing methods. Subsequently, the degree of complexity involved in this field drives researchers to continuously improve the existing feature selection and extraction techniques, highlighting the essential work that is necessary to propel the field forward. Unlike the existing reviews that focus on individual components such as preprocessing, feature selection, or feature extraction in isolation, this study offers a unified perspective by integrating



Type of Reviews

2- Systematic Literature Review

RQ1: How important is text preprocessing, and what are the key text preprocessing methods?

To address RQ1, this paper delves into the diverse landscape of text preprocessing techniques employed in text mining. The review spans various methods that can be applied to optimize textual data for subsequent analyses.

RQ2: What are the strengths and limitations of feature selection methods for text mining?

To answer RQ2, this survey examines the recent advancements in feature selection methods across various text domains. It provides a comparative analysis of various techniques, highlighting their strengths, limitations, and impacts on text-mining applications.

RQ3: How have feature extraction techniques evolved in the field of text mining (from traditional methods to advanced approaches), and how do they compare in terms of effectiveness?

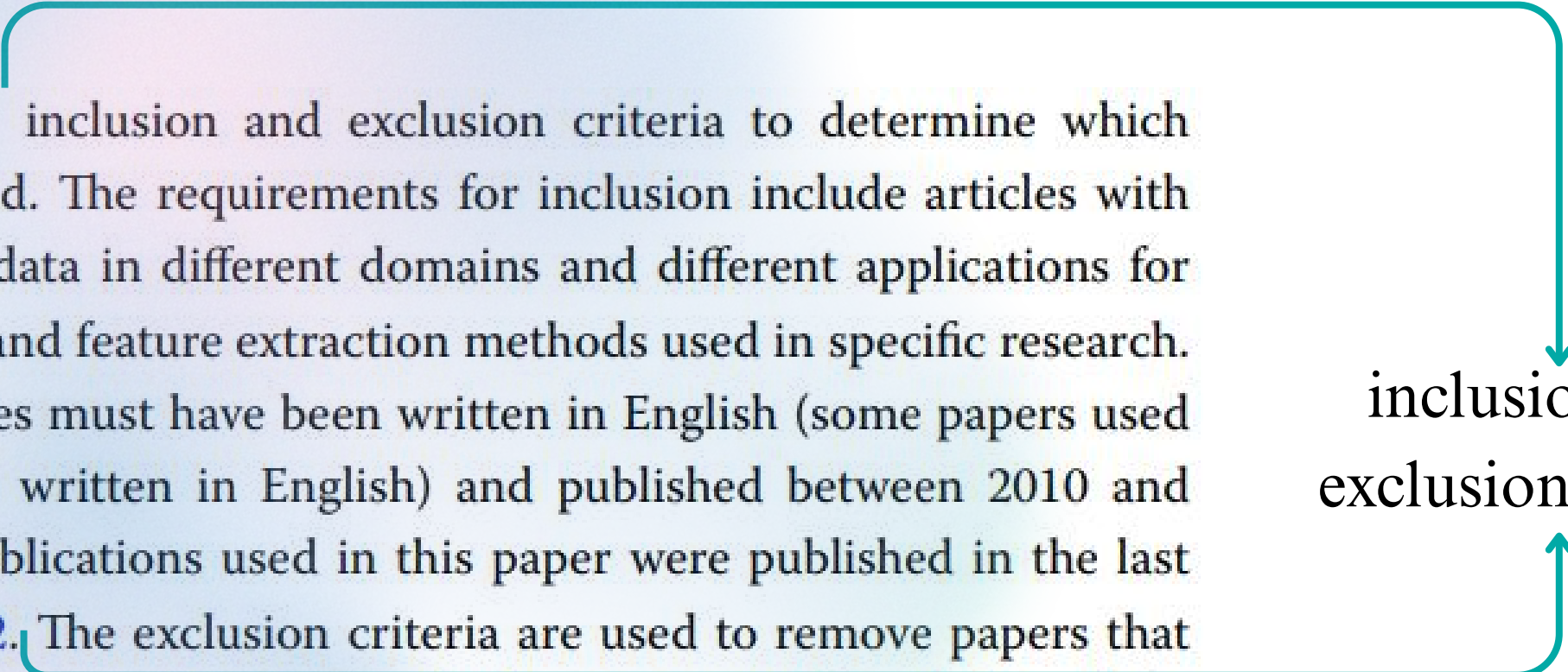
To answer RQ3, this study explores the most common feature extraction techniques that are currently prevalent in text mining. We analyze and summarize the recent trends involving feature extraction methodologies, considering their applications, strengths, and limitations.

Systematic
review
questions

Type of Reviews

2- Systematic Literature Review

This study adopts particular inclusion and exclusion criteria to determine which publications should be reviewed. The requirements for inclusion include articles with implications that used textual data in different domains and different applications for analyzing the feature selection and feature extraction methods used in specific research. Additionally, the selected articles must have been written in English (some papers used non-English datasets but were written in English) and published between 2010 and 2024. However, most of the publications used in this paper were published in the last five years, as shown in Figure 2. The exclusion criteria are used to remove papers that do not mention feature selection or extraction methods. To ensure the validity of the search methodology for the research questions related to text mining, feature selection, and feature extraction methods, a structured approach is followed, as illustrated in Figure 1. First, searches are conducted across multiple databases, including IEEE, MDPI,



inclusion and
exclusion criteria.

Type of Reviews

3. Scoping Review

A scoping review systematically maps the available literature on a broad or emerging topic. It identifies key concepts, research methods, evidence sources, knowledge gaps, and the overall scope of existing research.

- Best for: Exploring a broad, emerging, or under-researched area where the available evidence is diverse.

Type of Reviews

3. Scoping Review

REVIEW ARTICLE

Open Access

A scoping review on how generative artificial intelligence transforms assessment in higher education



Qi Xia³ , Xiaojing Weng² , Fan Ouyang³ , Tzung Jin Lin⁴ and Thomas K.F. Chiu^{1,5,6*} 

*Correspondence:
Thomas.kf.chiu@gmail.com;
tchiu@cuhk.edu.hk

¹ Department of Curriculum
and Instruction, The Chinese
University of Hong Kong, Hong
Kong, China

² Department of Curriculum
and Instruction, The Education

Abstract

Generative artificial intelligence provides both opportunities and challenges for higher education. Existing literature has not properly investigated how this technology would impact assessment in higher education. This scoping review took a forward-thinking approach to investigate how generative artificial intelligence transforms assessment in higher education. We used the PRISMA extension for scoping reviews to select articles for review and report the results. In the screening, we retrieved 969 articles and selected



Type of Reviews

4- Integrative Review

An integrative review combines and critically analyzes findings from different types of research, including quantitative, qualitative, and theoretical studies. It aims to develop a comprehensive understanding of a topic and may generate new theories, models, or conceptual frameworks.

➤ Best for: Synthesizing diverse evidence to build new perspectives, theories, or frameworks.

Type of Reviews

4- Integrative Review

Computers in Human Behavior 128 (2022) 107118



Contents lists available at [ScienceDirect](#)

Computers in Human Behavior

journal homepage: www.elsevier.com/locate/comphumbeh





The digital workplace and its dark side: An integrative review

Elizabeth Marsh ^{a,*}, Elvira Perez Vallejos ^b, Alexa Spence ^c

^a School of Psychology, University of Nottingham, University Park, Nottingham, NG7 2RD, United Kingdom
^b Associate Professor of Digital Technology and Mental Health, Faculty of Medicine & Health Sciences, University of Nottingham, United Kingdom
^c Associate Professor, School of Psychology, University of Nottingham, United Kingdom

ARTICLE INFO

Keywords:
Digital workplace
Technostress
Overload
Addiction
Integrative literature review

ABSTRACT

An intensification of digital working driven by Covid-19 has brought into sharp focus both the beneficial nature of digital workplace technologies and their potential dark side. Research has burgeoned in this area in recent years, but an integrated view across fields, technologies, dark side effects and outcomes is lacking. There are potential insights to be gained from compiling and comparing results and theoretical approaches. Following integrative review procedures, 194 studies were analysed to understand unintended negative consequences of a range of workplace technologies across disciplines and methodologies. The results demonstrate that considerable

Steps to a High-Quality Literature Review

Step 1: Define Research Scope

- Formulate a clear research question or aim.
- Identify keywords and concepts related to your topic.

Step 2: Search Systematically

- Use academic databases:
 - IEEE Xplore
 - ACM Digital Library
 - Scopus
 - Web of Science
 - Google Scholar
- Apply Boolean operators (AND, OR, NOT) to refine searches.

Steps to a High-Quality Literature Review

➤ How to Choose the Right Papers?

1. Indexing

- Ensure the paper is published in a journal indexed in reputable databases:

-  • Scopus
-  • Web of Science (ISI)
-  • Journal Citation Reports (JCR)
- IEEE Xplore, ACM DL, PubMed

Steps to a High-Quality Literature Review

➤ How to Choose the Right Papers?

2. Journal Impact

- Consider the Impact Factor (IF) from Journal Citation Reports (JCR).
- Alternatively:
- CiteScore (Scopus)
- SJR (SCImago Journal Rank)
- h-index (for authors or journals)

Steps to a High-Quality Literature Review

➤ How to Choose the Right Papers?

3. Paper-Level Quality

Publication metrics should not be used alone. Evaluate the paper's relevance, methodology, dataset, experimental design, limitations, and contribution to the research question.

Steps to a High-Quality Literature Review

➤ How to Choose the Right Papers?

4. Recency

- Prioritize papers from the last five years.
- Include older foundational studies when necessary.

Steps to a High-Quality Literature Review

➤ How to Choose the Right Papers?

5. Citation Influence

- Check citations using Google Scholar, Scopus, or Web of Science.
- High citation counts may indicate influence, not necessarily quality.
- Consider the publication year.
- Compare papers within the same field.

Steps to a High-Quality Literature Review

➤ How to Choose the Right Papers?

☐	Indexed	Published in recognized databases such as Scopus, Web of Science, IEEE Xplore, or PubMed.
☐	High Impact	Appears in a reputable journal or conference with strong academic influence.
☐	Relevant	Directly addresses the research topic, question, population, method, or context.
☐	Recent	Prefer studies published within the last five years, unless an older paper is foundational.
☐	Well-Cited	Has been frequently referenced by other researchers, indicating influence in the field.
☐	Peer-Reviewed	Evaluated by independent experts before publication.
☐	Methodologically Sound	Uses appropriate data, experimental design, evaluation metrics, and statistical analysis.
☐	Transparent	Clearly reports the methodology, limitations, datasets, and results.

Steps to a High-Quality Literature Review

Step 3: Organize and Compare the Selected Studies

Paper	Title & Authors	Aim/Research Question	Methodology	Key Findings	Limitations/Gaps	Relevance to My Study
Paper 1	Title, Author(s)	What does the study ask?	Qualitative/Quantitative	Main results	Weak sample size, no AI focus	Directly related to X
Paper 2	...	...	...	...	...	...

Steps to a High-Quality Literature Review

Step 3: Organize and Compare the Selected Studies

Paper	Assign a number or short label to each study.
Title & Authors	Write the paper title, authors, publication year, and venue.
Aim / Research Question	Summarize the main purpose or question of the study.
Methodology	Describe the research design, method, model, experiment, or analysis used.
Dataset / Sample	Record the dataset, participants, sample size, or data source.
Key Findings	Summarize the most important results or conclusions.
Limitations / Gaps	Identify weaknesses, missing evaluations, biases, or unresolved issues.

Steps to a High-Quality Literature Review

Step 4: Generate Insights from a Literature Matrix

➤ Identify Patterns

- Which findings are repeated across multiple studies?
- Are similar methods, datasets, variables, or evaluation metrics commonly used?
- Do the studies reach consistent conclusions?

Steps to a High-Quality Literature Review

Step 4: Generate Insights from a Literature Matrix

➤ Identify Gaps

- Which research questions remain unanswered?
- Which populations, datasets, methods, or contexts are underexplored?
- What limitations repeatedly appear in previous studies?

Steps to a High-Quality Literature Review

Step 4: Generate Insights from a Literature Matrix

➤ Trace Research Trends

- How has the field evolved over time?
- Which methods or topics are gaining attention?
- Which approaches are becoming less common or outdated?

Steps to a High-Quality Literature Review

Step 4: Generate Insights from a Literature Matrix

➤ Compare Contradictory Findings

- Do studies report different or conflicting results?
- What differences in data, methodology, or context might explain these contradictions?

Steps to a High-Quality Literature Review

Step 4: Generate Insights from a Literature Matrix

➤ Connect the Insights

- Which limitation will your research address?
- How will your study extend, improve, or challenge previous work?

Steps to a High-Quality Literature Review

Common Mistakes

- ✘ Reviewing only a small selection of papers and failing to cover the main studies in the field.
- ✘ Summarizing papers one by one, “Study 1, Study 2, Study 3”, without comparing or connecting their findings.
- ✘ Describing previous work without showing how it leads to the research gap and proposed study.

Activity 2

Join your project group and prepare the following:

- Select two relevant research paper related to your project topic.
- Complete one row of the literature review matrix.



Duration: 10 minutes

Literature Review matrix





GitHub Repo



Q&A

THANK YOU